Waste treatment plant Specification

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Revision	A
Status	APPROVED
Classification	INTERNAL
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Approved by	Selene Abaroth
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1. Purpose

This specification defines the performance, interface and verification requirements for the waste treatment plant (WM-WTP) of the Waste Management & Recycling on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-WM-03 (WBS 1.10.3), owned by Theron Wolcott.

The prime supplier for flight hardware is Brentaal Logistics Guild. Responsible engineer: Treva Bresk.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-WM-DWG-0202	Waste treatment plant general arrangement drawing
SHV-WM-BOM-0302	Waste treatment plant bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-LS-WM	Interface Control Document: Life Support & Atmospherics / Waste Management & Recycling

4. Description

Waste management collects, compacts, treats and recycles station waste through compactor units, treatment plants and a materials recycling line.

The waste treatment plant is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
WM-WTP-R01	The waste treatment plant shall provide a throughput of 1,951 t/day.	1,951 t/day	±1%	Test

5.1 General requirements

ID	Requirement	Verification
WM-WTP-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
WM-WTP-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
WM-WTP-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
WM-WTP-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
WM-WTP-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the WM test-report series.

8. Revision history

Rev	Date	Description	Author
A	2022-09-20	Initial release.	Treva Bresk